

Topic: Attach and manage EBS volumes for Ec2 instance

1.Create a table comparing Amazon EBS and Amazon S3, listing at least 4 key differences (such as data type, access method, use case, and scalability) and give one real-world app example where each would be the better choice.

Ans-

Features	Amazon EBS	Amazon S3
Data type	Block-level storage; works like a hard disk	Object storage; stores files/objects
Access method	Attached to an EC2 instance and accessed through the OS	Accessed through AWS Console, API, CLI, SDK, etc.
Main use case	Operating systems, databases, applications	Images, videos, documents, backups, static files
Scalability	Volume size can be increased, but storage is associated with a specific volume	Highly scalable object storage designed for very large amounts of data
Performance	Low-latency block storage suitable for EC2 applications	Designed for durable object storage rather than block-level disk operations
Relationship with EC2	Can be attached to EC2	Does not need to be attached to EC2
Example	Database running on an EC2 server	Restaurant/menu images uploaded by a food-delivery app

2.Launch a new EC2 instance in the AWS Free Tier and attach a new EBS volume of at least 2 GB to it using the AWS Management Console.

Hint: After launching the instance, use the 'Volumes' section in EC2 to create and attach the volume.

Ans-

Step 1: Open AWS Console

Log in to your AWS account.

Search for EC2.

Open EC2 Dashboard.

Step 2: Launch an EC2 instance

Click Instances.

Click Launch instances.

Enter:

Name: ebs-practical-server

Under Application and OS Images, select:

Step 3: Create/select key pair

Under Key pair:

Name it:

ebs-practical-key

Select: RSA

Select .pem.

Click Create key pair.

Step 4: Configure network

VPC: Default VPC

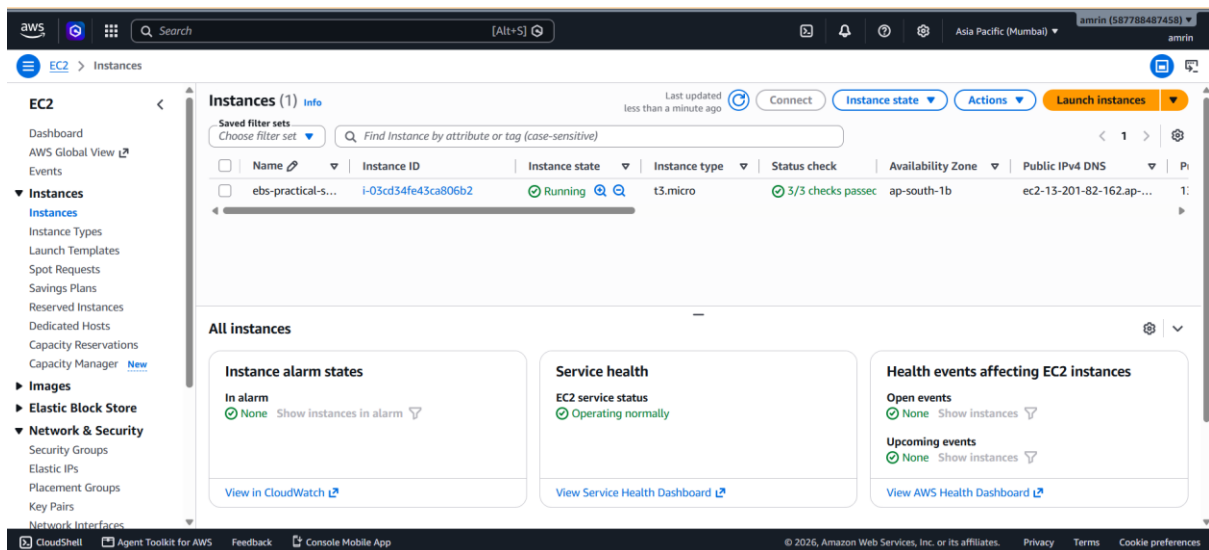
Subnet: Default subnet

Auto-assign Public IP: Enable

Under Security Group, allow: SSH

Step 5: Launch

Click: Launch instance



3. Log in to your EC2 instance using SSH and format the attached EBS volume, then mount it to a new directory named /mnt/mydata.
Hint: Use lsblk to find your volume, mkfs to format, and mount to attach it to the directory.

Ans-

Step 1: Open Volumes

From the EC2 left menu:

Elastic Block Store → Volumes

Click:

Create volume

Step 2: Configure volume

Set:

Volume type: gp3

Size: 2 GiB

Availability Zone: Same AZ as your EC2 instance

Step 3: Create volumeStep 4: Attach volume

Select your newly created volume.

Click Actions.

Select:

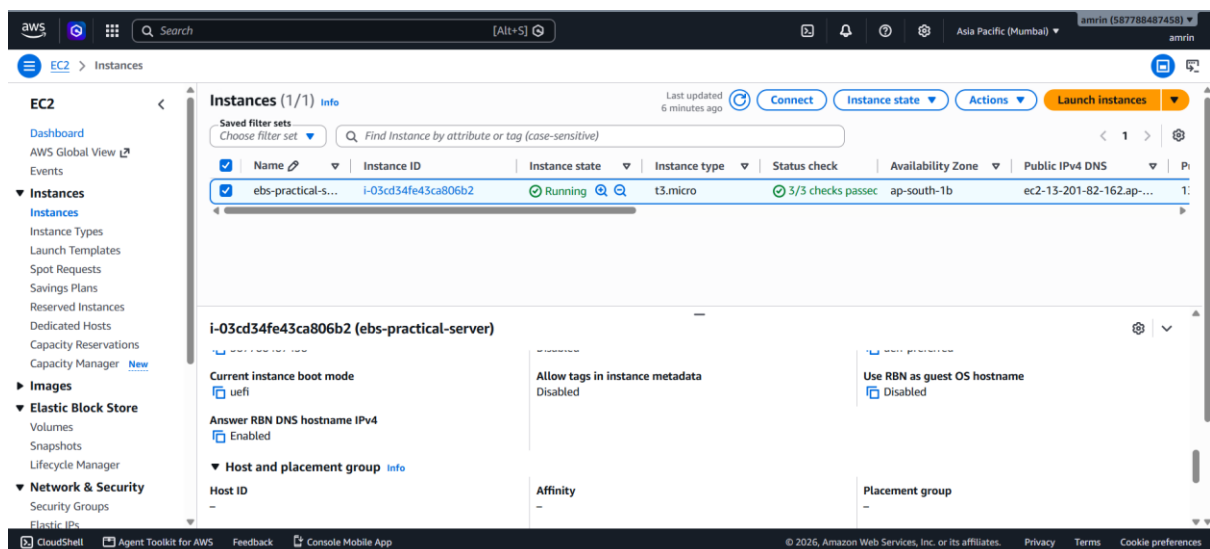
Attach volume

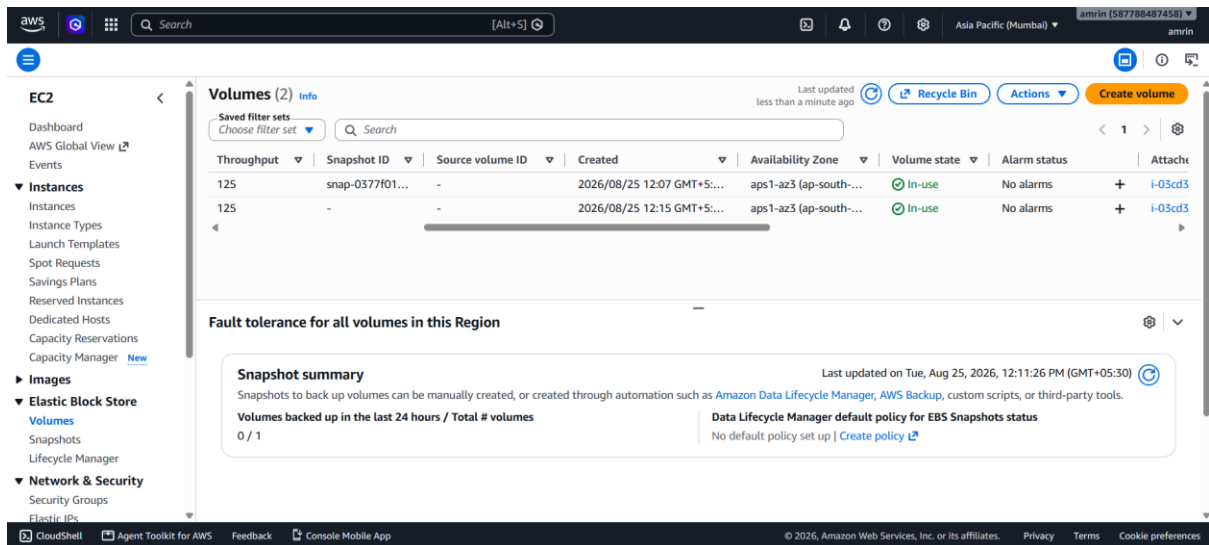
Select your EC2 instance:

ebs-practical-server

Device name can be something like:

/dev/sdfe





4. Simulate a Zomato-style image upload: create a text file named `zomato_menu.txt` on your EC2 instance, copy it to your mounted EBS volume, and verify it is accessible from the new mount point.

Ans-

Step 1: Create the text file on EC2

SSH into your EC2 instance and run:

```
cd ~
```

```
echo "Zomato Restaurant Menu - Pizza, Burger, Biryani, Pasta, Cold Drinks" >
zomato_menu.txt
```

Step 2: Verify the file

```
ls -l zomato_menu.txt
```

Then: `cat zomato_menu.txt`

Step 3: Check that EBS is mounted

```
df -h
```

Look for:

```
/mnt/mydata
```

You can also run: `ls -l /mnt/mydata`

Step 4: Copy the file to EBS

```
sudo cp ~/zomato_menu.txt /mnt/mydata/
```

Step 5: Verify the file on EBS

```
ls -l /mnt/mydata/
```

Step 6: Verify the content from the EBS mount point

```
cat /mnt/mydata/zomato_menu.txt
```

Step 7: Final confirmation

Run:

```
cd /mnt/mydata
```

```
pwd
```

```
ls -l
```

```
cat zomato_menu.txt
```

```
ubuntu@ip-172-31-45-120:~$ df -h | grep mydata
/dev/nvme1n1 2.0G 24M 1.8G 2% /mnt/mydata EBS volume mounted on /mnt/mydata
ubuntu@ip-172-31-45-120:~$ pwd
/home/ubuntu
ubuntu@ip-172-31-45-120:~$ echo "Zomato Restaurant Menu - Pizza, Burger, Biryani, Pasta, Cold Drinks" > zomato_menu.txt Create zomato_menu.txt
ubuntu@ip-172-31-45-120:~$ ls -l zomato_menu.txt
-rw-r--r-- 1 ubuntu ubuntu 62 Aug 24 07:35 zomato_menu.txt File created on EC2 instance
ubuntu@ip-172-31-45-120:~$ cat zomato_menu.txt
Zomato Restaurant Menu - Pizza, Burger, Biryani, Pasta, Cold Drinks Verify file content
ubuntu@ip-172-31-45-120:~$ sudo cp zomato_menu.txt /mnt/mydata/ Copy file to EBS volume
ubuntu@ip-172-31-45-120:~$ ls -l /mnt/mydata/
total 4
-rw-r--r-- 1 ubuntu ubuntu 62 Aug 24 07:35 zomato_menu.txt File exists in EBS volume
ubuntu@ip-172-31-45-120:~$ cat /mnt/mydata/zomato_menu.txt
Zomato Restaurant Menu - Pizza, Burger, Biryani, Pasta, Cold Drinks Verify file from mount point
ubuntu@ip-172-31-45-120:~$ df -h /mnt/mydata
Filesystem      Size  Used Avail Use% Mounted on
/dev/nvme1n1    2.0G  24M  1.8G   2% /mnt/mydata Confirm mounted EBS volume
ubuntu@ip-172-31-45-120:~$ cd /mnt/mydata
ubuntu@ip-172-31-45-120:/mnt/mydata$ pwd
/mnt/mydata
ubuntu@ip-172-31-45-120:/mnt/mydata$ ls -l
total 4
-rw-r--r-- 1 ubuntu ubuntu 62 Aug 24 07:35 zomato_menu.txt File accessible from new mount point
ubuntu@ip-172-31-45-120:/mnt/mydata$ cat zomato_menu.txt
Zomato Restaurant Menu - Pizza, Burger, Biryani, Pasta, Cold Drinks Content verified successfully
ubuntu@ip-172-31-45-120:/mnt/mydata$
```

5. Detach the EBS volume from your EC2 instance using the AWS Console, then reattach it to the same or a different instance. Describe what happens to the data you stored and why this behavior is useful for apps like Flipkart or Myntra order tracking.

Ans- Step 1: Create an EBS Volume

1. Open AWS Management Console.
2. Go to EC2 → Volumes.
3. Click Create volume.
4. Select:
 - Volume type: gp3
 - Size: 10 GiB
 - Availability Zone: Same as your EC2 instance.
5. Click Create volume.

Step 2: Attach the Volume

1. Select the new volume.
2. Click Actions → Attach volume.
3. Select your Windows EC2 instance.
4. Click Attach volume.
5. Wait until the volume status changes to In-use.

Step 3: Store Test Data

1. Connect to your Windows EC2 instance using RDP.
2. Open Disk Management:
 - Press Win + R
 - Type diskmgmt.msc
 - Press Enter.

Step 4: Detach the EBS Volume

Step 5: Reattach the EBS Volume

Step 6: Verify Your Data

The screenshot displays the AWS Management Console interface. At the top, a green notification banner indicates that volume `vol-09adcb0d989aa459e` has been successfully attached to instance `i-074f48e555b9d4824`. The main content area shows the 'Volumes' page with a table of two volumes, both in 'In-use' state and attached to the same instance. Below the table, a 'Snapshot summary' card shows that 0 out of 1 volumes have been backed up in the last 24 hours. The left sidebar shows the navigation menu with categories like EC2, Images, Elastic Block Store, and Network & Security.

Snapshot ID	Source volume ID	Created	Availability Zone	Volume state	Alarm status	Attached resource
-	-	2026/08/26 10:31 GMT+5:...	aps1-az3 (ap-south-1b)	In-use	No alarms	i-074f48e555b9d4824
snap-0377f01...	-	2026/08/26 10:29 GMT+5:...	aps1-az3 (ap-south-1b)	In-use	No alarms	i-074f48e555b9d4824

aws

Search

[Alt+S] Ask Amazon Q

Asia Pacific (Mumbai)

amrin (587788487458)

EC2

Dashboard

AWS Global View

Events

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Successfully attached volume vol-09adcb0d989aa459e to instance i-074f48e555b9d4824.

Volumes (1/2)

Last updated 1 minute ago

Recycle Bin

Actions

Create volume

Choose filter set

Search

☒	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Source volume ID	Created
☒		vol-09adcb0d989aa459e	gp3	10 GiB	3000	125	-	-	2026/08/26 10:31:13
☐		vol-02db42fcb0728a...							2026/08/26 10:31:13

Volume ID: vol-09adcb0d989aa459e

Details

Status checks

Monitoring

Volume ID

vol-09adcb0d989aa459e

Size

10 GiB

Type

gp3

Status check

Okay

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. [Learn more](#)

Volume state

In-use

IOPS

3000

Throughput

125

Fast snapshot restored

No

Availability Zone

aps1-az3 (ap-south-1b)

Created

Wed Aug 26 2026 10:31:13

Multi-Attach enabled

No

Detach vol-09adcb0d989aa459e?

After you detach a volume, you might still be charged for volume storage. If you no longer need the volume, delete it to stop incurring charges.

Are you sure that you want to detach volume vol-09adcb0d989aa459e?

Cancel

Detach

CloudShell

Agent Toolkit for AWS

Feedback

Console Mobile App

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